Programming for Problem Solving USING C (NES101/NES201)

Course Outcomes:

1. Proficiency in C Programming: Develop a solid foundation in C programming language, including syntax, data structures, and algorithms.
2. Problem-Solving Skills Enhancement: Improve problem-solving abilities by applying C programming concepts to solve real-world computational challenges efficiently.

Unit	Topic
I	**(Introduction to Programming) Introduction** to components of a computer system: Memory, processor, I/O Devices, storage, operating system, Concept of assembler, compiler, interpreter, loader and linker. Idea of Algorithm: Representation of Algorithm, Flowchart, and Pseudo code with examples, from algorithms to programs, source code. Programming Basics: Structure of C program, writing and executing the first C program, Syntax and logical errors in compilation, object and executable code. Components of C language. Standard I/O in C, Fundamental data types, Variables and memory locations, Storage classes.
II	**(Arithmetic expressions & Conditional Branching)** Arithmetic expressions and precedence: Operators and expression using numeric and relational operators, mixed operands, type conversion, logical operators, bit operations, assignment operator, operator precedence and associativity. Conditional Branching: Applying if and switch statements, nesting if and else, use of break and default with switch.
III	**(Loops & Functions)** Iteration and loops: use of while, do while and for loops, multiple loop variables, use of break and continue statements. Functions: Introduction, types of functions, functions with array, passing parameters to functions, call by value, call by reference, recursive functions.
IV	**(Arrays & Basic Algorithms)** Arrays: Array notation and representation, manipulating array elements, using multi-dimensional arrays. Character arrays and strings, Structure, union, enumerated data types, Array of structures, passing arrays to functions. Basic Algorithms: Searching &Basic Sorting Algorithms (Bubble, Insertion and Selection), Finding roots of equations, Notion of order of complexity.
V	**(Pointer & File Handling)** Pointers: Introduction, declaration, applications, Introduction to dynamic memory allocation (malloc, calloc, realloc, free), Use of pointers in self-referential structures, notion of linked list (no implementation) File handling: File I/O functions, Standard C pre-processors, defining and calling macros, command-line arguments.

TEXT BOOKS:

1. The C Programming Language" by Brian W. Kernighan and Dennis M. Ritchie
2. "C Programming: A Modern Approach" by K. N. King
3. "C How to Program" by Paul Deitel and Harvey Deitel
4. "C in a Nutshell" by Peter Prinz and Tony Crawford
5. "Let Us C" by Yashavant Kanetkar
6. "Problem Solving and Program Design in C" by Jeri R. Hanly and Elliot B. Koffman